1 Genome sequencing and assembly

Genomic DNA was extracted with the STE method. The harvested DNA was detected by the agarose gel electrophoresis and quantified by Qubit. The genome of XXX was sequenced by Single Molecule, Real-Time (SMRT) technology. Sequencing was performed at the Beijing Novogene Bioinformatics Technology Co., Ltd. The low quality reads were filtered by the SMRT Link v8.0 and the filtered reads were assemblied using software Canu to generate one contig without gaps.

Canu:(https://github.com/marbl/canu/,version：2.0)

2Genome Component prediction

Genome component prediction included the prediction of the coding gene, repetitive sequences, non-coding RNA, genomics islands, transposon, prophage, and clustered regularly interspaced short palindromic repeat sequences (CRISPR). The available steps were proceeded as follows: 1) For bacteria, we used the GeneMarkSprogram to retrieve the related coding gene. 2) The interspersed repetitive sequences were predicted using the RepeatMasker (http://www.repeatmasker.org/). The tandem Repeats were analyzed by the TRF (Tandem repeats finder). 3) Transfer RNA (tRNA) genes were predicted by the tRNAscan-SE . Ribosome RNA (rRNA) genes were analyzed by the rRNAmmer .Small nuclear RNAs(snRNA)were predicted by BLAST against the Rfam database. 4) The IslandPath-DIOMB program was used to predict the Genomics Islands, and transposon PSI was used to predict the transposons based on the homologous blast method. The PHAST was used for the prophage prediction(http://phast.wishartlab.com/) and the CRISPRFinder was used for the CRISPR identification.

3 Gene function

We used seven databases to predict gene functions. They were respective GO (Gene Ontology), KEGG (Kyoto Encyclopedia of Genes and Genomes), COG (Clusters of Orthologous Groups), NR (Non-Redundant Protein Database databases), TCDB (Transporter Classification Database), and, Swiss-Prot . A whole genome Blast search (E-value less than 1e-5, minimal alignment length percentage larger than 40%) was performed against above seven databases. The secretory proteins were predicted by the Signal P database, and theprediction of Type I-VII proteins secreted by the pathogenic bacteria were based on the EffectiveT3 software. Meanwhile, we analyzed the secondary metabolism gene clusters by the antiSMASH .For pathogenic bacteria, we added the pathogenicity and drug resistance analyses. We used the PHI (Pathogen Host Interactions), VF, ARDB (Antibiotic Resistance Genes Database) to perform the above analyses.Carbohydrate-Active enzymes were predicted by the Carbohydrate-Active enZYmes Database .

4 Comparative genomics analysis

Comparative genomic analysis included the genomic synteny, the core genes and specific genes, gene family phylogenetic tree, SNP(Single Nucleotide Polymorphism), indel (insertion and deletion) and SV(Structural Variation) annotation, and genome visualization. 1)Genomic alignment between the sample genome and reference genome (or among more than two sample genomes) were performed using the MUMmer and LASTZ tools. Genomic synteny was analyzed based on the alignment results. 2) Core genes and specific genes were analyzed by the CD-HIT rapid clustering of similar proteins software with a threshold of 50% pairwise identity and 0.7 length difference cutoff in amino acid. Then the venn figure was drawn to show their relationships among the samples. 3) Gene family was constructed by the gene of XXX and XXX, using the multiple softwares: Blast was used to pairwise align all genes and eliminate the redundancy by solar and carried out gene family clustering treatment based on the alignment results with Hcluster_sg software. 4) The phylogenetic tree was constructed by the TreeBeST or PhyML and the setting of bootstraps was 1,000 with the orthologous genes. 5) SNP, indel and SV were found by the genomic alignment results among samples by the MUMmer and LASTZ we mentioned them before. 6)Genome overview was created by Circos to show the annotation information.